Spatial interactive multi-objective optimization for land-use allocation under uncertainty

1. Please download anaconda or conda (lighter version of anaconda) on your desktop.
2. Open anaconda_prompt within anaconda.
3. Navigate to the folder where you want the code: < cd /path/to/your/projects >
4. Clone the repository containing the implementation of robustness check within PI-NSGA-II in land use optimization via: < git clone https://github.com/aditisinghxd/robust_pinsga2_for_landuse_optimization.git>

   (When copy pasting in the terminal make sure the code is in one line)

5. Once cloned, navigate within the cloned folder via:

   < cd robust_pinsga2_for_landuse_optimization>

Creating an anaconda environment to run the code within

6. Create a new environment: < conda create -n robpinsga2env python=3.9 -y >

   You can also name the environment as suited:

   < conda create -n <env_name> python=3.9 -y >

7. Activate the environment: < conda activate robpinsga2env >
8. Clone the pymoo repository (containing PI-NSGA-II algorithm) via:

   < git clone https://github.com/Human-Centered-Optimization-Lab/pymoo-pi-emo.git >

9. Once cloned, navigate within the pymoo folder via: < cd pymoo-pi-emo >
10. Switch to the PI-NSGA-II branch via: < git checkout feature/pinsga2 >
11. To be done outside anaconda: replace the pinsga2 file within the folder pymoo-pi-emo\pymoo\algorithms\moo with the pinsga2 file stored within the folder update_pinsga_file_in_pymoo-pi-emo
12. Once done, continue within anaconda_promt: < pip install . >
13. Install these dependencies within the environment:

    < conda install ipykernel pandas seaborn -y >

14. Install ema_workbench within the environment:

    < pip install ema-workbench >

15. The environment is now set up. Navigate back to the main folder: < cd .. >
16. Just to make sure the kernel is visible in JupyterNotebook:

< python -m ipykernel install --user --name=robpinsga2env --display-name "Python (robpinsga2env)" >

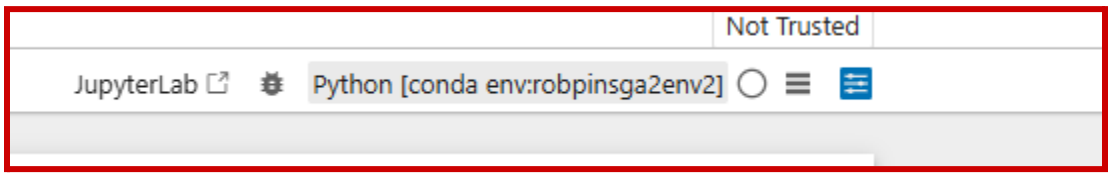
(When copy pasting in the terminal make sure the code is in one line)

17. Since the environment is already currently active, you can now go to jupyter notebook in the browser (jupyter notebook handles the GUI window better) by using: < jupyter notebook [path to your folder] > . For example: I used

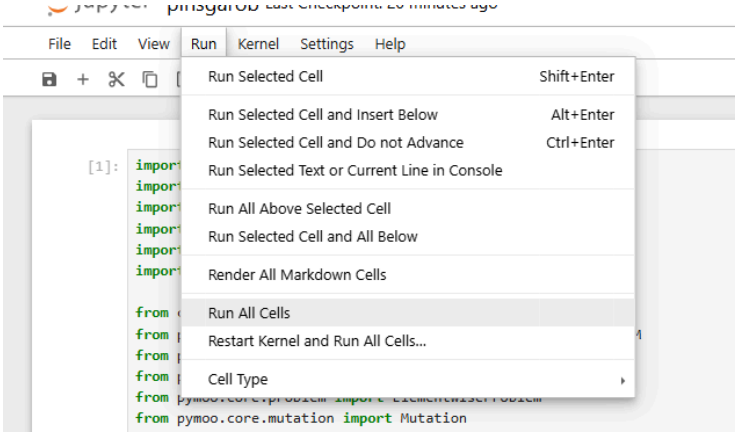
< jupyter notebook "C:\Users\marsh\Desktop\robust_pinsga2_for_landuse_optimization >

Here you have all the project files visible and you can edit them as you go.

18. Before running the model, you can adjust the experiment parameters in the config.ini file (e.g., map size, population size, uncertainty settings, etc.) according to your requirements.
19. Now in the jupyter notebook in the browser please open pinsgarob.ipynb file. On the top right select the environment you created for the project.



20. Then run the entire file by using: Run All Cells



Note: For future use, simply activate the environment (step 7) and go to the jupyter notebook via (step 16).